

1. The following command was executed to build Docker images for both the frontend (Node.js) and backend (Flask) services and to start the containers using Docker Compose. This ensures that all services are built from their respective Dockerfiles and run together in a shared Docker network.

```
C:\docker\node-flask-docker-app>docker-compose up --build
time="2025-12-25T15:14:45+05:30" level=warning msg="C:\\docker\\node-flask-docker-app\\docker-compose.yml: the attribute
'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Building 3.7s (25/25) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 1.06kB 0.0s
=> [backend internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 190B 0.0s
=> [frontend internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 156B 0.0s
=> [frontend internal] load metadata for docker.io/library/node:18 2.4s
=> [backend internal] load metadata for docker.io/library/python:3.10 2.4s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [frontend internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [backend internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [frontend 1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556 0.1s
=> => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783 0.1s
=> [frontend internal] load build context 0.0s
=> => transferring context: 251B 0.0s
=> [backend 1/5] FROM docker.io/library/python:3.10@sha256:ad84630de0e8b6f2ee92b4e2996b08c269efa96be13558d0233ed 0.1s
=> => resolve docker.io/library/python:3.10@sha256:ad84630de0e8b6f2ee92b4e2996b08c269efa96be13558d0233ed08a0e190 0.1s
=> [backend internal] load build context 0.0s
```

```
[+] Running 4/4
✔node-flask-docker-app-backend Built 0.0s
✔node-flask-docker-app-frontend Built 0.0s
✔Container node-flask-docker-app-backend-1 Recreated 0.4s
✔Container node-flask-docker-app-frontend-1 Recreated 0.3s
Attaching to backend-1, frontend-1
backend-1 | * Serving Flask app 'app'
backend-1 | * Debug mode: off
backend-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
backend-1 | * Running on all addresses (0.0.0.0)
backend-1 | * Running on http://127.0.0.1:5000
backend-1 | * Running on http://172.19.0.2:5000
backend-1 | Press CTRL+C to quit
frontend-1 |
frontend-1 | > frontend@1.0.0 start
frontend-1 | > node server.js
frontend-1 |
frontend-1 | Frontend running on port 3000
backend-1 | 172.19.0.1 -- [25/Dec/2025 09:45:22] "GET / HTTP/1.1" 404 -
backend-1 | 172.19.0.1 -- [25/Dec/2025 09:45:23] "GET / HTTP/1.1" 404 -
backend-1 | 172.19.0.3 -- [25/Dec/2025 09:46:00] "POST /api/signup HTTP/1.1" 200 -
```

2. The docker images command is used to verify that the Docker images for the frontend and backend applications were successfully created. The output confirms the image names, tags, image IDs, and sizes.

```
C:\docker\node-flask-docker-app>docker images
```

| IMAGE                                 | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|---------------------------------------|--------------|------------|--------------|-------|
| backend:latest                        | 6da23eb230   | 1.62GB     | 415MB        | U     |
| docker/welcome-to-docker:latest       | c4d56c24da4f | 22.2MB     | 6.03MB       |       |
| frontend:latest                       | f4beb9205690 | 1.59GB     | 400MB        | U     |
| node-flask-docker-app-backend:latest  | 8743ec199c3a | 1.6GB      | 411MB        | U     |
| node-flask-docker-app-frontend:latest | 9f2cbf7bbfb6 | 1.59GB     | 400MB        | U     |
| python-hello-world:latest             | e4763fae7cf3 | 183MB      | 45MB         |       |
| python:latest                         | 492b292a9449 | 1.63GB     | 432MB        |       |
| ubuntu:latest                         | c35e29c94501 | 119MB      | 31.7MB       |       |

```
C:\docker\node-flask-docker-app>docker network ls
```

| NETWORK ID   | NAME                          | DRIVER | SCOPE |
|--------------|-------------------------------|--------|-------|
| 3a84846bc544 | app-network                   | bridge | local |
| 178c9d078484 | bridge                        | bridge | local |
| 6a7c7a523e0c | host                          | host   | local |
| 1f1dd47f89ca | node-flask-docker-app_default | bridge | local |
| 4570087c0bc4 | none                          | null   | local |

- The frontend application was successfully accessed through a web browser using localhost:3000. This confirms that the Node.js frontend container is running correctly and is able to communicate with the Flask backend service.



